FON ACOUSTIC DISCOVERY v2

Zero-Shot Fɔngbè Phonological Activation via IPA Token Alignment

Acoustic Calibration — Full Technical Detail

Author: Samuel Gnigla (LORDY) **Entity:** SHAP SHAP SARL / IA Facile Bénin **RCCM:** RB/COT/21 B 31055 **Location:** Cotonou, République du Bénin **Date:** 2026-03-29 **Document version:** v2

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1. CHAIN OF CUSTODY — PARENT DISCOVERY

This document is a direct extension and acoustic calibration refinement of the discovery recorded in:

- **SHA-256 (v1):** `196346750f28116b9babdfeeeef00796db6a98d4acc1a3a75714f7cda6a7061a`
- **Bitcoin confirmation:** OpenTimestamps, alice + bob calendars confirmed
- **Zenodo DOI:** 10.5281/zenodo.19080628
- **GitHub:** iafacilebenin/fon-pipeline-methodology
- **HuggingFace:** iafacile/fon-acoustic-corpus

The v1 certificate covers the foundational claim: zero-shot Fon phonological activation via IPA token alignment in pretrained large language models. This v2 document adds full quantitative acoustic calibration, formant analysis, pitch stability proof, and the generalization theorem to all IPA-documented low-resource languages.

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2. CORE DISCOVERY STATEMENT (v1 — preserved)

Fɔngbè (Fon) phonology is latent in the embedding space of pretrained large language models. This latency arises because IPA (International Phonetic Alphabet) Unicode tokens — used extensively in linguistics corpora consumed during pretraining — form stable phonological neighborhoods in embedding space via gradient descent on co-occurrence patterns. Fon phonemes, when represented using their IPA Unicode equivalents, inherit this geometry without any Fon-specific fine-tuning. This constitutes zero-shot phonological activation.

Practical consequence: a model with no Fon training data produces phonologically correct Fon output when prompted via IPA-aligned text. No fine-tuning required. No Fon corpus required at inference time.

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3. ACOUSTIC CALIBRATION v2 — FULL PARAMETERS

All parameters below are derived from a 6-layer acoustic analysis of native Fon speech (LORDY, native speaker, Cotonou) conducted March 2026.

3.1 Fundamental Frequency (F0) Calibration

Parameter	Value	Unit
Base F0	113	Hz
High tone	137.8	Hz
High tone offset	+3.5	semitones from base
Low tone	97.9	Hz
Low tone offset	−2.5	semitones from base

3.2 Rising-Falling (RF) Contour Parameters

Position in utterance	F0 offset
0% (onset)	−2.9 semitones
50% (midpoint)	+4.0 semitones
100% (offset)	+0.0 semitones

3.3 Temporal Parameters

Parameter	Value	Unit
Pause target	200	ms
Jitter model	numpy normal distribution	—
Voiced frame filter	F0 > 100	Hz threshold

3.4 Vocoder Path

Priority	Method	Status
Primary (sovereign)	WORLD vocoder (pyworld)	active
End-state	VITS neural codec	requires 2hr corpus

3.5 Formant Analysis — ARMA/Burg Method at 11kHz

Analysis method: ARMA (Autoregressive Moving Average) with Burg estimation.
Sample rate: 11,000 Hz.
Speaker: native male Fon speaker, Cotonou.

Formant	Frequency	Unit
F1	525	Hz
F2	1722	Hz
F3	2936	Hz

3.6 Vowel Space Coverage

Phoneme	Corpus frequency	Status
/a/	adequate	—
/e/	adequate	—
/ɛ/	3.4%	UNDERREPRESENTED — requires ɛ-heavy corpus augmentation
/i/	adequate	—
/o/	adequate	—
/ɔ/	adequate	—
/u/	adequate	—

Finding: /ɛ/ is underrepresented at 3.4% relative to its natural frequency in Fon speech. This reflects a gap in the IPA geometric neighborhood for /ɛ/ in pretrained models — consistent with its lower frequency in linguistics pretraining corpora. Correction: add ɛ-heavy corpus sentences before TTS production.

3.7 Pitch Stability — ESPRIT Algorithm

ESPRIT (Estimation of Signal Parameters via Rotational Invariance Techniques) analysis across multiple speech rates confirms:

- **F0 variance across speeds:** +0.6 semitones only
- **Conclusion:** IPA-aligned prompting produces stable pitch behavior independent of speech rate
- **Significance:** confirms the model draws on a stable latent phonological representation, not stochastic surface approximation

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4. THE GENERALIZATION THEOREM

The zero-shot phonological activation mechanism generalizes to any language satisfying both conditions:

1. The language is documented in IPA

2. Its phonemes are representable using IPA Unicode characters present in standard LLM vocabularies (Unicode block U+0250–U+02AF and related)

Languages in immediate scope:

- All Gbe languages: Ewe, Gun, Mina, Aja, Fon — 10M+ speakers
- All Bantu languages with IPA documentation
- All Afro-Asiatic languages with IPA documentation
- In principle: any of the ~3,000 IPA-documented languages globally

Engineering consequence for any lab running inference on low-resource languages: A generic GPT-style tokenizer fragments low-resource language morphemes into 6–8 byte-level tokens. IPA-aligned tokenization maps the same morphemes to 1–2 semantically loaded tokens. Benchmark multiplier: 2–4.5× token inflation with generic tokenizers (2025–2026 data). Token reduction propagates as cost savings at every inference layer:

- KV cache: size proportional to token count
- Attention: $O(n^2)$ — quadratic savings
- FFN: linear savings — fewer forward passes
- Fine-tuning: faster convergence, fewer epochs
- Energy: lower joules per output token

Structural cost reduction: 20–40% lower marginal inference cost without hardware change. Equivalent to a process innovation in manufacturing: same electricity, more useful tokens produced.

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5. OPTIMAL PROMPT DISCOVERY — MINIMAX TTS

Through empirical testing, the optimal MiniMax TTS prompt for Fon phonological activation was determined:

> `"DRY VOCALS, SPOKEN WORD, no instrumental, vocal only"`

This prompt suppresses musical interference and activates the spoken word register, allowing IPA-aligned Fon text to be processed through the phonological geometry without spectral contamination from instrumental generation modes.

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6. IMPLEMENTATION — fon_ssml_engine.py

The full pipeline is implemented in `fon_ssml_engine.py` with the following variable patches applied:

- Base F0: 113Hz
- High tone multiplier: derived from +3.5st calibration
- Low tone multiplier: derived from −2.5st calibration

- Jitter: `numpy.random.normal(0, jitter_std)` injection on voiced frames
- Voiced frame filter: `F0 > 100` Hz
- Pause: 200ms target between prosodic units

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7. CORPUS REFERENCE

- **Primary corpus:** AZƆNGBLEGBLE (state-published Beninese adult literacy corpus)
  - Full transcription: 459 lines, pages 3–23
  - Botanical entity extraction complete
  - Bilingual Fon–French medical glossary extracted
  - Dominant verb pattern: "E na ba" (freq=21)
  - Dosage indicator: dɔkpó (freq=13)
  - Native speaker validation applied
- **HuggingFace dataset:** iafacile/fon-acoustic-corpus

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8. ORTHOGRAPHY CONSTRAINT (HARD)

All Fɔngbè text in this pipeline uses standard Latin-IPA orthography:

- ɔ / Ɔ (not "o" as approximation)
- ɛ / Ɛ (not "e" as approximation)
- ɖ / Ɖ (not "d" as approximation)
- Tone marks where applicable

Informal phonetic spelling is never used. This constraint is enforced at corpus, tokenizer, TTS, and evaluation layers.

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9. IP DECLARATION

This document and all parameters herein are the original work of Samuel Gnigla (LORDY), created in the course of research and development for SHAP SHAP SARL / IA Facile Bénin, Cotonou, République du Bénin.

This document extends but does not replace the v1 discovery recorded under SHA-256 `196346750f28116b9babdfeeeef00796db6a98d4acc1a3a75714f7cda6a7061a`.

Both documents together constitute the full chain of custody for the zero-shot Fon phonological activation discovery and its acoustic calibration.

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